

Algebra II

Course Outline • Mathlify Summer Program

Course Overview

This course develops advanced algebraic thinking through a functions-based and transformational approach. Students strengthen algebraic fluency, modeling skills, and problem-solving strategies required for Precalculus and AP-level mathematics.

Course Units

Unit 1 — Foundations & Functions

- Function notation, domain and range
- Evaluating and interpreting functions
- Transformations of functions

Unit 2 — Absolute Value

- Absolute value functions
- Absolute value equations and inequalities

Unit 3 — Quadratic Functions & Equations

- Graphing quadratics in multiple forms
- Solving quadratic equations and inequalities
- Quadratic systems and relations

Unit 4 — Polynomial Functions

- Polynomial behavior and end behavior
- Factoring and polynomial equations

Unit 5 — Rational Functions & Expressions

- Rational expressions and restrictions
- Solving rational equations

Unit 6 — Radical Functions

- Radical expressions and equations
- Rational exponents

Unit 7 — Exponential & Logarithmic Functions

- Exponential growth and decay
- Modeling real-world situations
- Logarithmic properties
- Solving exponential and logarithmic equations